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[SMART HURL]

by

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ABSTRACT

The Smart Hurl project aims to develop a fully embedded system capable of detecting ball strike events on a hurl during gameplay using onboard sensing and edge based processing. This interim report presents the design and validation of a real time motion classification framework implemented on the Arduino Nano 33 BLE Sense Rev2.

Accelerometer data from the integrated BMI270 inertial measurement unit were sampled at 50 Hz and processed locally to compute orientation invariant acceleration magnitude features. Feature extraction was performed using fixed, non overlapping 50 sample windows 1 second duration. For each window, the mean and standard deviation of acceleration magnitude were calculated and used to train a logistic regression classifier offline. The trained model parameters were algebraically converted for direct deployment on the microcontroller, enabling real time inference without runtime feature scaling.

A controlled walking versus idle classification task was used to validate the embedded machine learning workflow. A total of 127 labelled windows were extracted, and statistical analysis demonstrated complete linear separability between classes using the standard deviation feature. The classifier achieved 100% training accuracy on the collected dataset.

The system architecture is organized into sensing, processing, and communication layers, with Bluetooth Low Energy used to stream classification outputs to external devices. The results demonstrate that real time edge based motion classification is achievable on a resource constrained microcontroller, establishing a validated embedded framework for future extension toward hurl specific strike detection and impact analysis.

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I am also grateful to Ben Toland for his assistance in the early stages of the project, particularly for providing insights that contributed to the decision to adopt the Arduino Nano 33 BLE Sense as a single integrated platform, rather than pursuing the initial dual sensor approach.

DECLARATION

The work submitted in this report is the results of the candidate's own investigations and has not been submitted for any other award. Where use has been made of the work of other people it has been fully acknowledged and referenced.

Student Name:

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LIST OF ABBREVIATIONS

IMU	Inertial Measurement Unit
BLE	Bluetooth Low Energy
ML	Machine Learning
BMI	Bosch Motion Sensor
JSON	JavaScript Object Notation
Hz	Hertz
W	Weight
B	Bias
MCU	Microcontroller
DFT	Discrete Fourier Transform

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Chapter 1 Introduction

1.1 Problem

Hurling is a high speed striking sport in which performance quality is largely assessed subjectively by players and coaches. Unlike larger global sports markets, hurling has limited integration of embedded sensing technologies capable of providing objective, quantitative feedback during gameplay.

The primary technical challenge in developing a smart hurl system lies in reliably detecting ball strike events using onboard sensors under strict embedded constraints. A strike generates a short duration, high-magnitude acceleration event that must be distinguished from normal motion patterns such as swinging, running, or handling the hurley. This requires robust signal processing and decision making within the limitations of a battery powered microcontroller platform. Many existing smart sports systems rely on cloud based processing or external sensing infrastructure. However, continuous wireless connectivity increases latency, power consumption, and system complexity. For a compact, embedded device mounted directly on a hurley, real time edge based detection is essential.

The central problem addressed in this project is therefore whether reliable ball strike detection can be achieved using onboard inertial sensing and real time embedded processing without reliance on continuous cloud connectivity.

1.2 Background

Advances in microcontroller performance, low power wireless communication, and miniaturized inertial sensors have enabled the development of intelligent embedded systems for sports analytics. Modern microcontrollers are capable of performing lightweight machine learning inference directly on device, allowing sensor data to be processed locally without continuous reliance on cloud infrastructure.

Inertial measurement units incorporating tri-axial accelerometers and gyroscopes, are widely used for motion analysis in wearable and sports equipment applications. These sensors are capable of capturing dynamic movement characteristics, including acceleration peaks, vibration signatures, and rapid orientation changes. When combined with suitable signal processing techniques, inertial data can be used to identify specific motion events in real time.

Edge based processing has emerged as an important architectural approach for embedded systems operating under power and connectivity constraints. By performing feature extraction and classification locally, systems can reduce communication overhead, minimise latency, and operate independently of stable internet access. This approach is particularly suitable for sports equipment deployed in dynamic outdoor environments.

The Arduino Nano 33 BLE Sense Rev2 integrates an IMU, Bluetooth Low Energy communication, and sufficient processing capability to support real time embedded inference. These characteristics make it an appropriate platform for investigating onboard motion classification as a foundation for strike detection in a smart hurl system.

1.3 Scope and Objectives

This project's scope is confined to designing and testing a smart hurl prototype that uses embedded sensors and on device computing to identify ball striking events. The work focuses on gathering data from inertial sensors, recognizing distinctive hit signals and figuring out whether reliable strike detection is possible given the limitations of an embedded battery powered system.

Real time coaching feedback, player talent evaluation and in depth biomechanical analysis is deemed outside the purview of this research. Furthermore, although the idea of determining the ideal impact location also known as the "sweet spot" is intriguing, its execution is viewed as possible future work rather than the main goal of the project.

The walking versus idle classification task is used solely as a validation framework for the embedded inference pipeline.

1.4 Document Overview

This report begins by introducing the Smart Hurley project, which aims to develop an embedded system capable of detecting strikes on a hurley during gameplay. The background section establishes the importance of motion classification in sports performance monitoring and outlines the technologies that enable this, including the Arduino Nano 33 BLE Sense Rev2 microcontroller, the integrated BMI270 inertial measurement unit, Bluetooth Low Energy communication, and cloud based database frameworks. These components form the technical foundation for implementing and testing embedded machine learning classifiers on a resource constrained platform.

The analysis and design section details the work completed to date, focusing on demonstrating the capabilities of the Nano 33 BLE Sense for real time classification. Section 3.2 describes the implementation of a walking and idle classifier using accelerometer data, feature extraction, and a logistic regression model. This task was used as a controlled demonstration of the embedded system's ability to acquire sensor data, compute orientation invariant features, and perform classification in real time. Section 3.3 presents the system architecture, showing how sensing, processing, and communication layers interact to produce a fully embedded data pipeline, from IMU acquisition to classification output.

Sections 3.4 and 3.5 summarize progress and outline the plan for remaining work.

Achievements include successful deployment of the walking/idle classifier, BLE streaming for real time monitoring, and preliminary cloud database integration. Limitations observed during testing and the next steps required to extend the system toward its ultimate objective of detecting hurl strikes are also discussed. Collectively, these sections provide a clear overview of both the progress made and the planned future development of the Smart Hurl project.

Chapter 2 Literature Review

This chapter reviews the current state of the art in smart sports equipment, impact detection, and embedded sensing systems relevant to the development of a smart hurl. The aim of the review is to place the project within the context of existing scientific and technical work, and to justify the design choices made in later chapters.

The review begins by considering the broader use of embedded sensors and inertial measurement units in sports performance analysis, with particular emphasis on striking and impact based applications. The benefits and drawbacks of current methods for impact detection, signal characterization, and event classification are discussed. Because it directly affects system architecture and implementation choices, the usage of embedded and edge based processing in resource constrained systems is also highlighted.

The choice of the devices, data processing procedures and system boundaries used in this study are justified by the appropriate theoretical ideas, sensing techniques, and experimental approaches found in the literature.

2.1 Smart Sports Equipment and Wearable Sensing

In recent years embedded sensors have been integrated into sports equipment to provide objective quantitative feedback on performance, expanding beyond simple fitness trackers to equipment specific analytics. A notable example is the use of smart sensors in tennis rackets, where similar challenges of impact detection and motion analysis exist.

Commercial tennis sensor systems such as the Sony Smart Tennis Sensor and Babolat Pure Drive Play integrate three axis motion sensors and vibration sensors into the racket handle or attachable modules. By measuring shot type (forehand, backhand, serve, smash), swing speed, ball speed, ball spin, and impact location on the racket strings, these devices accurately determine the location of the ball strike and the quality of the hit. These technologies can classify strokes and give players comprehensive analytics on stroke mechanics, which were previously assessed subjectively by coaches. [1]

These commercial trends are mirrored in research efforts. To determine stroke quality in real time, for example embedded systems that integrate IMUs with extra sensors and algorithms for learning have been developed. These systems quantify characteristics including shot power, racket face angle and the point of contact on the racket. These systems typically

process sensor data either on a connected device or locally on a microcontroller, highlighting both the potential and challenges of embedding analytics directly into sports gear. [2]

Studies using wearable IMU systems in tennis have also demonstrated the feasibility of recognizing distinct stroke actions (e.g. forehand, backhand) by placing sensors on the player's body or equipment and applying classification algorithms. Such research shows that inertial data can reliably capture dynamic motion events in striking sports and can be used to distinguish between different types of movements relevant to performance analysis.[3]

These tennis examples show how embedded sensor technologies which detect not just general motion but also impact events and mechanical characteristics of individual strikes can transform conventional sports equipment into analytical performance tools. The creation of a smart hurl system, which uses comparable inertial sensing techniques to identify ball strikes and possibly evaluate strike quality, can benefit from this body of work. Similar to how certain smart tennis technologies determine impact location on the racket, future implementations might also consider features like sweet spot detection based on impact signal patterns.

2.2 Impact Detection Using Inertial Sensors

In the area of smart sports equipment impact detection is a major focus especially in striking sports where the time and quality of contact between the tool and the ball are crucial to performance. An impact causes a high magnitude short duration reaction that spreads throughout the apparatus as vibration and abrupt acceleration changes. Because of these features inertial sensors in particular accelerometers are ideal for identifying impact occurrences.

Based on earlier studies impact events can frequently be recognized by examining acceleration magnitude peaks that above typical motion levels connected to handling or swinging motions. Due to their low computational complexity and suitability for embedded systems with limited processing power, straightforward threshold based techniques are frequently employed to differentiate impacts from non impact movements [4]. These methods are frequently documented in sports like baseball and tennis, where hits produce specific acceleration signatures that deviate from typical movement patterns. [5]

The literature has also examined more sophisticated methods such as machine learning based classification and frequency domain analysis. While machine learning techniques can be trained to identify intricate patterns linked to various hit kinds frequency based approaches take advantage of the fact that impacts introduce higher frequency components compared to

smooth human motion [2]. These methods can increase detection accuracy, but they usually need more memory, processing power, and labelled datasets, which may make them less appropriate for low power embedded systems. [4]

The literature also emphasizes how a few variables such as strike force, contact position, equipment material, and sensor placement affect impact signals. Because of this impact signals can differ greatly even within the same sport making generalized detection techniques difficult [4]. This variation highlights the necessity of application specific assessment when modifying impact detection techniques for new gear or athletic environments.

Overall existing research shows that inertial sensors can accurately identify impact events in striking sports, but it also shows that system limitations and algorithm complexity need to be carefully considered. These results guide the method used in this study which evaluates impact detection techniques in the setting of a hurl and applies them within the constraints of an embedded system.

2.3 Embedded Edge Based Data Processing

Instead of sending raw data to an external system for processing embedded and edge based data processing analyses sensor data directly on the device where it is collected. This method is frequently applied in smart sports equipment to minimize wireless data transfer lower latency and increase power efficiency all of which are crucial limitations for battery powered embedded systems. [6]

Microcontrollers are used by many smart sporting gadgets to handle inertial sensor data in real time, allowing for the quick recognition of events like collisions, strokes, or motion phases. Instead of transmitting continuous raw sensor streams, these systems can selectively broadcast just the necessary information, like impact timestamps or summary statistics, and react immediately to recognized events by processing data locally. This method prolongs the device's functioning life and drastically lowers communication overhead.

The literature highlights that algorithm selection for embedded processing must balance detection accuracy with computational complexity. Simple methods, such as threshold based impact detection or time domain feature extraction, are commonly favored due to their low memory usage and deterministic execution time [4]. While more complex techniques, including machine learning models, have been demonstrated on embedded platforms, they often require careful optimisation and may not be suitable for low power systems without hardware acceleration [6].

Also, edge based computing increases system reliability by allowing operation without constant wireless access. This is especially important for sporting goods used in training or outdoor settings, where reliable data transfer cannot be ensured [2]. Even in situations where communication is momentarily unavailable, local processing makes sure that important events are identified and recorded.

Chapter 3 Implementation

3.1 Hardware Design

The Smart Hurl system's hardware design was developed through an iterative approach that was influenced by benchmarking against current smart sports equipment and weekly evaluations. Early design choices concentrated on collecting ball hurley impact events and motion characteristics while preserving a small, sturdy embedded device appropriate for usage in an athletic setting.

3.1.1 Initial Sensor Selection

The Smart Hurl system was initially designed as a dual sensor architecture with a piezoelectric sensor for direct impact detection and an IMU for motion tracking. The BMI160 IMU was selected for its six degrees of freedom, suitable for capturing swing speed, rotational motion, and overall hurley movement. The Murata 7BB-20-6L0 piezoelectric sensor was intended to detect strikes with high sensitivity, providing an analog signal proportional to impact force.

During early implementation, the Arduino Nano 33 BLE Sense was adopted as a single integrated solution. This board includes the BMI270 IMU, providing tri-axial accelerometer and gyroscope measurements without requiring a separate sensor. Using the Nano 33 BLE Sense simplified the hardware setup, reduced wiring complexity, and ensured consistent data acquisition across all experiments. It also provides a sufficiently powerful microcontroller to perform real time feature extraction and logistic regression classification on device, which was essential for demonstrating embedded ML workflows with walking and idle data. Additionally, its built in BLE capabilities allow real time streaming of classification outputs to external devices, supporting monitoring and future app integration.

3.1.2 Wireless Communication and Connectivity

To simplify future data transfer and system extension, wireless connectivity was taken into consideration. Due to being suitable for low power short range embedded systems, Bluetooth communication was given precedence over Wi-Fi.

Although Wi-Fi provides direct internet access and high data transmission, it usually uses more power and requires constant network association. For a small, battery operated sports device designed to function in dynamic settings, these features are undesirable. Bluetooth Low Energy, on the other hand is better suited for embedded sensing applications since it is

optimized for irregular communication, low energy consumption, and short range functioning.

Microcontrollers that enable Bluetooth 4.2 and Bluetooth 5.0 were taken into consideration throughout the platform evaluation. While Bluetooth 4.2 offers dependable low power communication for simple data transfer, Bluetooth 5.0 offers notable enhancements, such as a 2 Mbit/s low energy physical layer, enhanced robustness in noisy radio settings, and an extended communication range. Thanks to these improvements, Bluetooth 5.0 is now more suitable for sending event based sensor data while using less power.

The Arduino Nano 33 BLE Sense Rev2 and ESP32 based platforms both include built in Bluetooth functionality, which reduces total hardware complexity and does away with the need for external wireless modules.

3.1.3 Microcontroller Platform Evaluation and Selection

During the project a number of microcontroller platforms were assessed, but ESP32 based development boards like the ESP32 DevKit V1 and FireBeetle C6 were given attention. For cloud connected IoT applications these devices provide flexibility, integrated wireless connectivity and high processing performance.

High clock speeds and dual core processing are two benefits of the ESP32 architecture that make it appropriate for applications requiring network based processing or frequent wireless connectivity. Nevertheless, a number of restrictions were noted for the Smart Hurl application. ESP32 based systems typically require external sensors to be interfaced via I²C or SPI, increasing wiring complexity, physical footprint, and potential points of mechanical failure.

The Arduino Nano 33 BLE Sense Rev2, on the other hand provided a more comprehensive solution that complied with the project specifications. This platform eliminates the need for an external inertial sensor by utilizing an integrated IMU, Bluetooth Low Energy radio, and low power microcontroller. By lowering the number of interconnections this greatly simplified the hardware design and increased robustness.

Strong interaction with the Arduino environment which includes well maintained libraries, extensive documentation and support for quick prototyping, is advantageous from a development standpoint for the Arduino Nano 33 BLE Sense Rev2. When compared to handling several separate components on an ESP32 based system, these factors decreased development time and integration risk. Power efficiency was a further consideration. The Arduino Nano 33 BLE Sense Rev2 is designed for low power operation and is well suited to

edge based processing, where continuous sensing and local computation are required. While ESP32 platforms remain a viable option for future system iterations involving increased wireless communication or cloud connectivity, the Arduino platform was better suited to the current project focus on local impact detection and system simplicity.

3.1.4 Benchmarking Against Existing Smart Sports Equipment

Existing smart sports equipment was analysed especially in the tennis domain to validate the chosen sensing approach. IMUs with accelerometers and gyroscopes are used in commercial systems like Babolat Play and the Zepp Tennis Sensor to monitor swing speed, racket orientation and stroke parameters. These systems show how inertial sensing can be used to capture useful performance metrics in striking sports.

Instead of using specialized force sensors these systems also use impact detection methods based on vibration and acceleration analysis. This strategy was reflected in the original Smart Hurl design, which combined a piezoelectric sensor and IMU based motion tracking to offer a direct reaction to ball strikes.

3.1.5 Design Evolution

Though the original dual sensor hardware approach offered extensive sensing capabilities more research revealed real world difficulties with piezoelectric impact sensing. These included the need for further analog signal conditioning, signal instability and sensitivity to mounting conditions. These elements decreased resilience and increased system complexity, especially in a high impact athletic setting. To directly detect impact events the sensing strategy changed to a more integrated inertial sensing approach, using high rate acceleration data from the onboard IMU. This kept enough sensitivity to pick up short duration impact signatures while lowering hardware complexity.

The Arduino Nano 33 BLE Sense Rev2's embedded machine learning capable microcontroller was another element that influenced the final hardware direction. The board is ideal for edge based inference tasks since it has a microprocessor built to handle on device machine learning workloads. The availability of specialized ML support was seen as an appealing feature that gave room for future improvement, even though machine learning is not a fundamental necessity of the current system implementation. This feature also presented a new technological hurdle which pushed the project to go beyond threshold based detection and toward more sophisticated data driven methods in the next stages of development. This is

in line with the project's overarching goal of investigating intelligent embedded systems while maintaining a reliable and functional baseline implementation.

3.2 Software Implementation

This project used a simulation driven approach to software implementation instead of instantly deploying complex logic onto embedded hardware. Before committing to immediate on device deployment, this approach made it possible to investigate, test, and improve the Smart Hurl system's behaviour using recorded and fake sensor data. This method guaranteed that ensuing hardware based testing was based on verified hypotheses, decreased development risk and facilitated quick iteration.

3.2.1 Offline Behaviour Simulation Using Synthetic Data

Offline behavior simulation was used as an early stage validation method to explore expected system responses before committing to continuous hardware testing. A Python based simulation script was developed to generate synthetic IMU data including three axis accelerometer and gyroscope which were written to a raw data file the generated signals were designed to approximate characteristic motion patterns such as low level handling, rapid swing acceleration and abrupt impact like events.

This synthetic dataset enabled rapid iteration of signal processing logic, including magnitude calculation and basic smoothing operations, without being constrained by sensor availability, battery life, or experimental repeatability. By analyzing the simulated data offline it was possible to verify that distinct motion intensities produced measurably different signal profiles and that impact style events resulted in identifiable peaks.

The use of synthetic data at this stage was not intended to replace real sensor measurements, but rather to confirm feasibility and guide coming embedded development. The outcomes of this simulation phase directly informed later firmware work on the arduino nano.

3.2.2 Embedded Data Acquisition

Following offline simulation, firmware was developed for the Arduino Nano 33 BLE Sense Rev2 to acquire real world inertial sensor data and validate the feasibility of on device processing. This stage focused on confirming that the embedded platform could reliably sample accelerometer data, apply basic signal conditioning, and perform simple classification in real time.

The implemented firmware reads three axis acceleration data from the onboard IMU and computes the overall acceleration magnitude to provide a single representative measure of movement intensity. To reduce the influence of short term noise and sensor jitter, a moving average filter was applied to the magnitude signal before further processing.

Using the smooth acceleration magnitude, threshold based logic was implemented to classify system behavior into stationary or walking. While these activity labels are not the final targets of the Smart Hurl system, they were intentionally chosen as a controlled validation task. This allowed the reliability of sensor sampling, filtering, and decision logic to be assessed under varying motion conditions without introducing unnecessary complexity. Classification results were printed to the serial monitor in real time, providing immediate feedback during testing and enabling rapid verification of system behavior. This embedded implementation confirmed that the Nano 33 BLE Sense Rev2 can perform continuous sensor acquisition and basic decision making within the constraints of an embedded environment, thereby establishing a solid foundation for later impact specific detection and more advanced processing stages.

3.2.3 Real Time On Device Logistic Regression Classification

Following initial experimentation with single feature classification, a refined machine learning pipeline was implemented to ensure robust and defensible real time embedded inference on the Arduino Nano 33 BLE Sense Rev2. The objective was to perform reliable walking versus idle detection using statistically validated features and properly trained model parameters.

Raw Data Acquisition:

Raw accelerometer data (ax, ay, az) were streamed over the serial interface at 50 Hz and recorded directly to CSV files using a Python serial listener. Two datasets were collected:

- Idle (label = 0)
- Walking (label = 1)

To eliminate orientation dependence, the acceleration magnitude was computed:

$$|a| = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

This transformation produces a single orientation invariant scalar feature representing overall movement intensity.

Feature extraction was performed using fixed, non overlapping 50 sample windows, corresponding to a duration of approximately 1 second per window at 50 Hz. For each window, two statistical features were computed:

- Mean acceleration magnitude (mag_mean)
- Standard deviation of acceleration magnitude (mag_std)

Each window was assigned the corresponding activity label, resulting in a structured feature dataset for supervised learning.

Statistical Separation

The extracted feature ranges were:

idle = 77 windows mag_std range = 0.00053 – 0.00281

walking = 50 windows mag_std range = 0.0697 – 0.1624

No overlap was observed between the two classes in the standard deviation feature, indicating clear separability.

Feature scaling and model training

Because the extracted features operate on different numeric scales (mag_mean = 1.0, mag_std = 0.001–0.16), feature standardisation was applied prior to training. The feature vectors were normalised using the training dataset mean and standard deviation to ensure balanced parameter estimation.

A logistic regression classifier was then trained on the scaled features to model the probability of walking versus idle activity.

3.2.4 Future Integration

With the walking/idle detection successfully deployed on the Arduino Nano 33 BLE Sense, the framework for real time embedded classification has been established. This workflow demonstrates that orientation independent features such as acceleration magnitude, combined with lightweight logistic regression models can be reliably executed on a resource constrained microcontroller. The next phase of the Smart Hurl project will extend this approach to hurling specific motion detection. This includes identifying swing patterns, strike events and potentially the sweet spot of contact with the sliotar.

3.3 System Architecture

The smart hurl system is implemented as a compact embedded architecture centred around the Arduino Nano 33 BLE Sense Rev2. This platform was selected due to its integrated sensing capabilities, onboard BLE communication, and sufficient computational resources to execute lightweight machine learning algorithms in real time.

The overall architecture is organised into three functional layers: sensing, processing, and communication. This layered structure ensures clear separation of responsibilities while maintaining efficient edge based execution.

Sensing layer

Motion data is acquired using the integrated BMI270 inertial measurement unit. Although the IMU provides both accelerometer and gyroscope measurements, the current implementation utilises accelerometer data only for activity classification. Tri-axial acceleration values (a_x , a_y , a_z) are sampled at a fixed rate of 50 Hz. These raw sensor measurements are passed directly to the processing layer for feature computation.

Processing layer

All feature extraction and classification operations are performed locally on the microcontroller. To reduce orientation dependence, the three acceleration components are transformed into an orientation invariant scalar by computing the acceleration magnitude:

$$|a| = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

This representation ensures robustness under varying hurler orientations during motion. During model development, fixed, non overlapping 50 sample windows 1second duration were used offline to extract statistical features. The trained logistic regression model parameters were then algebraically converted for direct deployment on the microcontroller.

On device inference is performed continuously using the trained weight and bias parameters. The classifier computes a probabilistic output, which is compared against a threshold ($P \geq 0.5$) to determine the activity state.

To reduce transient misclassification caused by short duration acceleration spikes, a short prediction smoothing stage is applied to the classifier output. This improves temporal stability without introducing significant computational overhead.

All sensing, feature transformation, and inference occur locally on the device, with no reliance on external or cloud based processing.

Communication layer

The communication layer enables wireless transmission of classification results via Bluetooth Low Energy. During testing and validation, classification outputs are streamed to an external device such as a laptop or smartphone application. This allows real time monitoring of system behavior without wired serial communication and supports future integration with a companion mobile application.

The complete layered system architecture is illustrated in Figure 1.

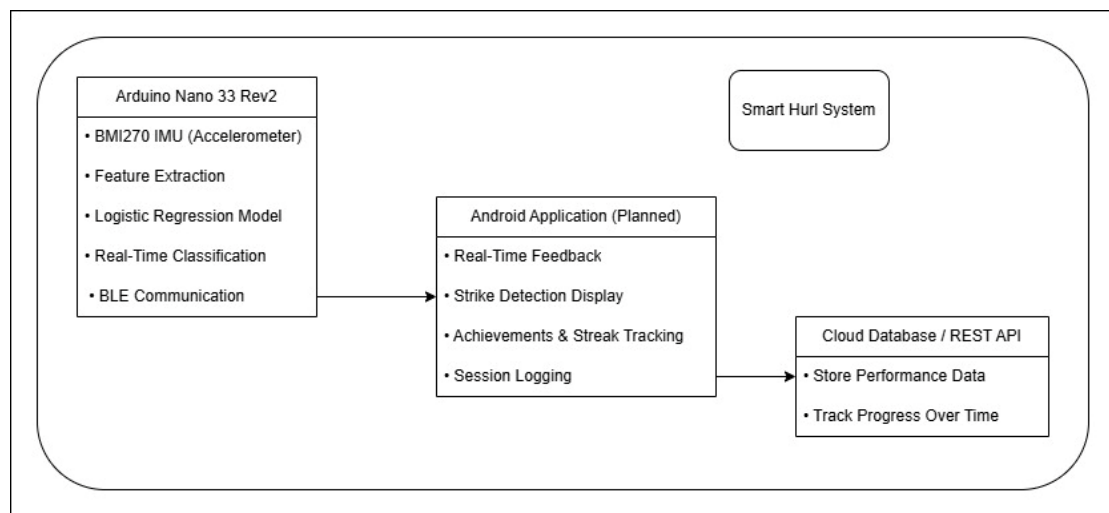


Figure 1 Smart Hurl Architecture

3.4 Progress & Work Done

The Smart Hurl project has progressed to a stage where multiple components of the embedded motion classification system have been successfully implemented and validated. The core of the system is the Arduino Nano 33 BLE Sense Rev2, incorporating the integrated BMI270 inertial measurement unit. The embedded platform is capable of acquiring accelerometer data in real time, performing local feature transformation, executing machine learning inference, and transmitting results wirelessly to an external device.

Real time walking versus idle detection has been successfully deployed on the Arduino Nano 33 BLE Sense. Tri-axial accelerometer readings are sampled at fixed intervals, and the orientation invariant acceleration magnitude is computed to provide a robust scalar representation of motion intensity. The computed feature is passed through a logistic regression model trained offline and converted for direct embedded execution. The model produces a probabilistic output, which is thresholded at 0.5 to determine the predicted activity state. To improve temporal stability and reduce sensitivity to short duration acceleration spikes, a short moving average filter is applied to the prediction output. This smoothing stage reduces transient classification fluctuations while maintaining real time responsiveness.

Testing under controlled conditions demonstrated that the system can reliably distinguish walking from idle activities. Minor instability was observed primarily during transitions between motion states, which is expected due to abrupt changes in acceleration characteristics.

Bluetooth Low Energy (BLE) functionality was implemented to enable real time wireless transmission of classification outputs. The Arduino Nano 33 BLE Sense operates as a BLE peripheral device, advertising a custom service to allow external devices to subscribe to activity prediction updates.

A custom BLE service was defined using the UUID:

19b10000-e8f2-537e-4f6c-d104768a1214

Within this service, a characteristic was configured to transmit classification results. The characteristic supports read and notify properties, enabling connected central devices (e.g., laptop or smartphone) to receive real time updates without continuous polling.

During testing, the Nano 33 BLE Sense successfully advertised its presence and was discoverable via its unique BLE MAC address. Connection stability was validated under both

USB powered and battery powered operation. Activity predictions were streamed reliably to a central device, confirming correct service registration, characteristic updates, and notification handling.

Database and Cloud Preparation

Preliminary groundwork has been completed for event based data logging to a cloud database. Classification events are structured in JSON format for upload to a remote database system. While full integration is ongoing, this establishes a scalable architecture for long term data storage, performance analysis, and future expansion toward strike detection analytics.

3.5 Future work

Building upon the progress achieved to date, the next phase of the Smart Hurl project will focus on extending the system's capabilities and performing comprehensive validation. The system will be expanded to include hurling specific motions. This includes collecting and labelling data for swing patterns, strike events, and detection of the sweet spot when contacting the sliotar. New features and models will be developed to classify these more complex motion types while maintaining real time performance on the embedded platform.

The embedded classification will be further enhanced. Logistic regression may be refined or replaced with more sophisticated models capable of handling multiple motion classes. Sampling rates and temporal segmentation strategies will be optimized to reduce latency and improve responsiveness.

Comprehensive real world testing will be conducted. The system will be evaluated under varying hurley orientations, dynamic user movements, and different environmental conditions. The results will inform adjustments to feature extraction, classification thresholds, and smoothing parameters to ensure reliable performance. BLE communication and cloud integration will be improved. The system will support reliable streaming of classification outputs and long term data logging to a cloud database. This will enable remote monitoring, performance analysis, and potential mobile application interfaces for users.

A companion mobile application will be developed to interface with the embedded system via BLE. The app will provide real time visualization of motion events and classify strikes, as well as track user performance metrics over time. Features such as achievements and streaks

will be included to provide feedback and motivation, supporting the overall goal of monitoring and improving hurley gameplay.

3.6 Experimental Evaluation

Sensor data was collected at a fixed sampling rate of 50 Hz. Feature extraction was performed using fixed, non overlapping 50 sample windows (1 second duration).

Parameter	Value
Sampling rate	50 Hz
Window size	50 samples
Window duration	1.0s
Idle windows	77
Walking windows	50
Total windows	127

Table 1 Feature Extraction

The standard deviation of acceleration magnitude (mag_std) showed clear class separation

Class	Mag_std Range
Idle	0.00053 – 0.00281
Walking	0.0697 – 0.1624

Table 2 Class Separation

Actual/Predicted	Idle	Walking
Idle	77	0
Walking	0	50

Table 3 Confusion Matrix

Training accuracy

The logistic regression classifier achieved 100% as seen in Table 3. As the dataset is linearly separable and limited in size, the perfect training accuracy reflects feature separability rather than model complexity. However, no independent test dataset was used at this stage.

Interpretation of results:

The absence of feature overlap in mag_std explains the perfect classification accuracy observed during training. The dataset is linearly separable, making logistic regression an appropriate and computationally efficient model choice for embedded deployment.

However, it should be noted that the dataset was collected under controlled conditions with limited environmental variation. While the results demonstrate feasibility, further testing under more diverse motion patterns is required to validate generalisation performance.

The dataset consisted of a single participant under controlled conditions. While the results demonstrate feasibility, additional data collection under varied real world scenarios is required to evaluate generalization performance.

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